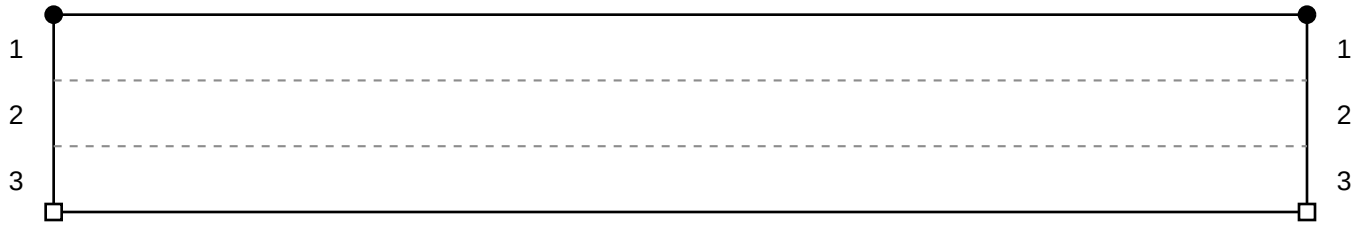
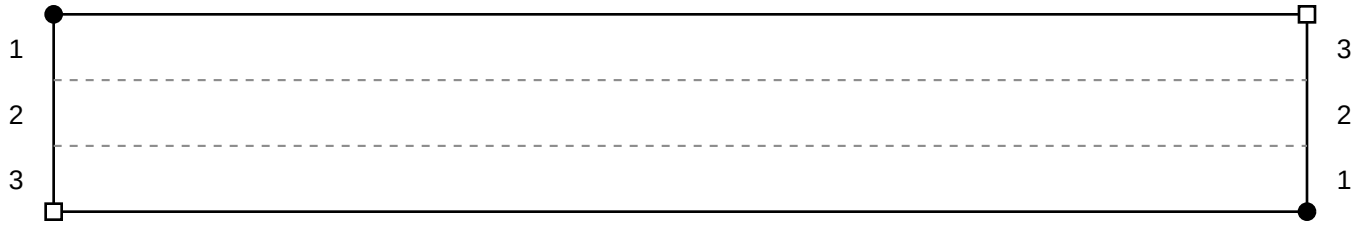
Keep bands on the table. An adult makes every cut. Rectangles are joining recipes: join matching corner marks. M matches the end order; R reverses it. A closed edge trip stops on its first return to its start.

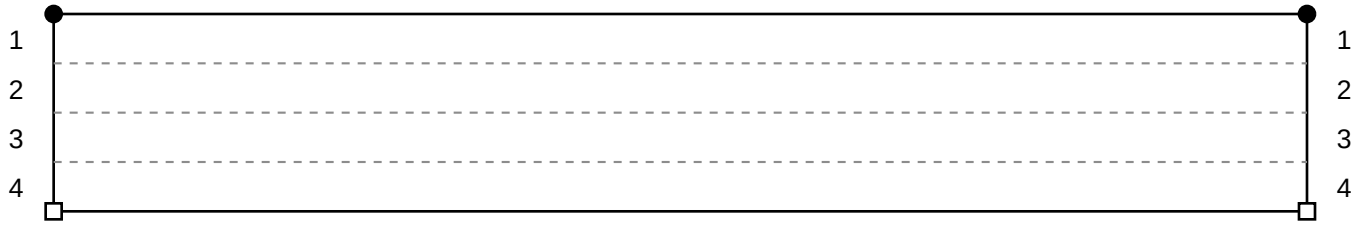
Problem 1: Make all four bands. Predict how many connected pieces the dotted cuts will make. Have an adult cut every lane boundary, then trace all the closed edges on each piece. Can one cut result contain both a one-edge piece and a two-edge piece?



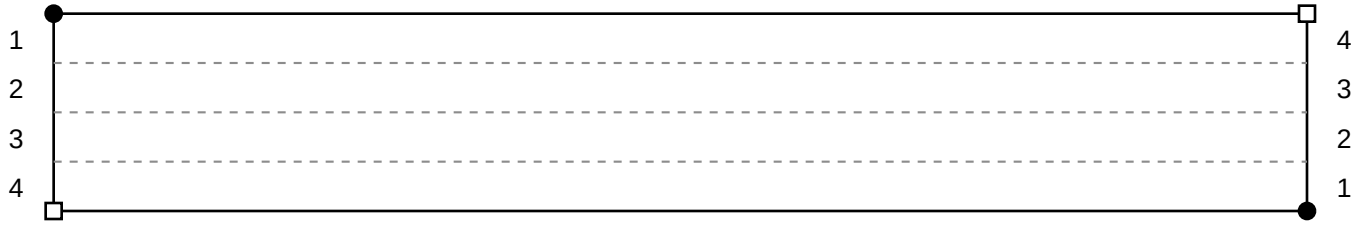
3 lanes / M



3 lanes / R

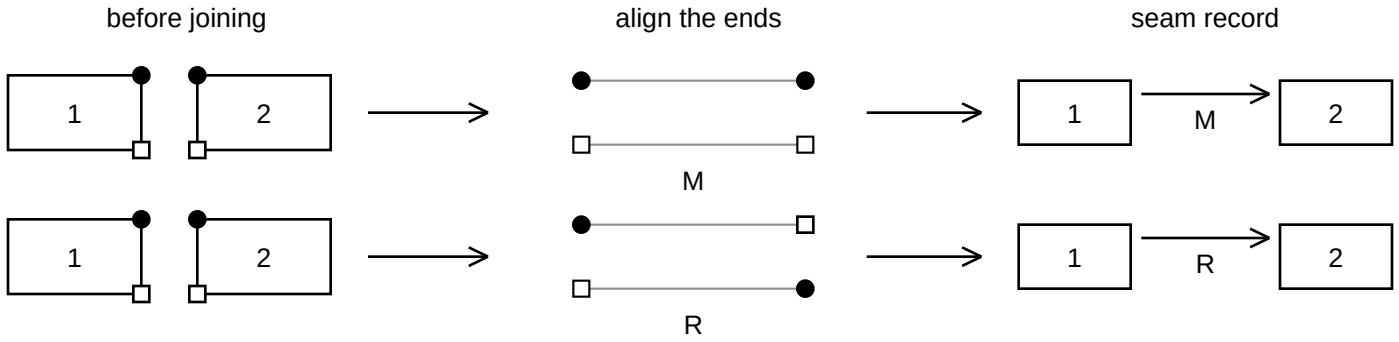


4 lanes / M

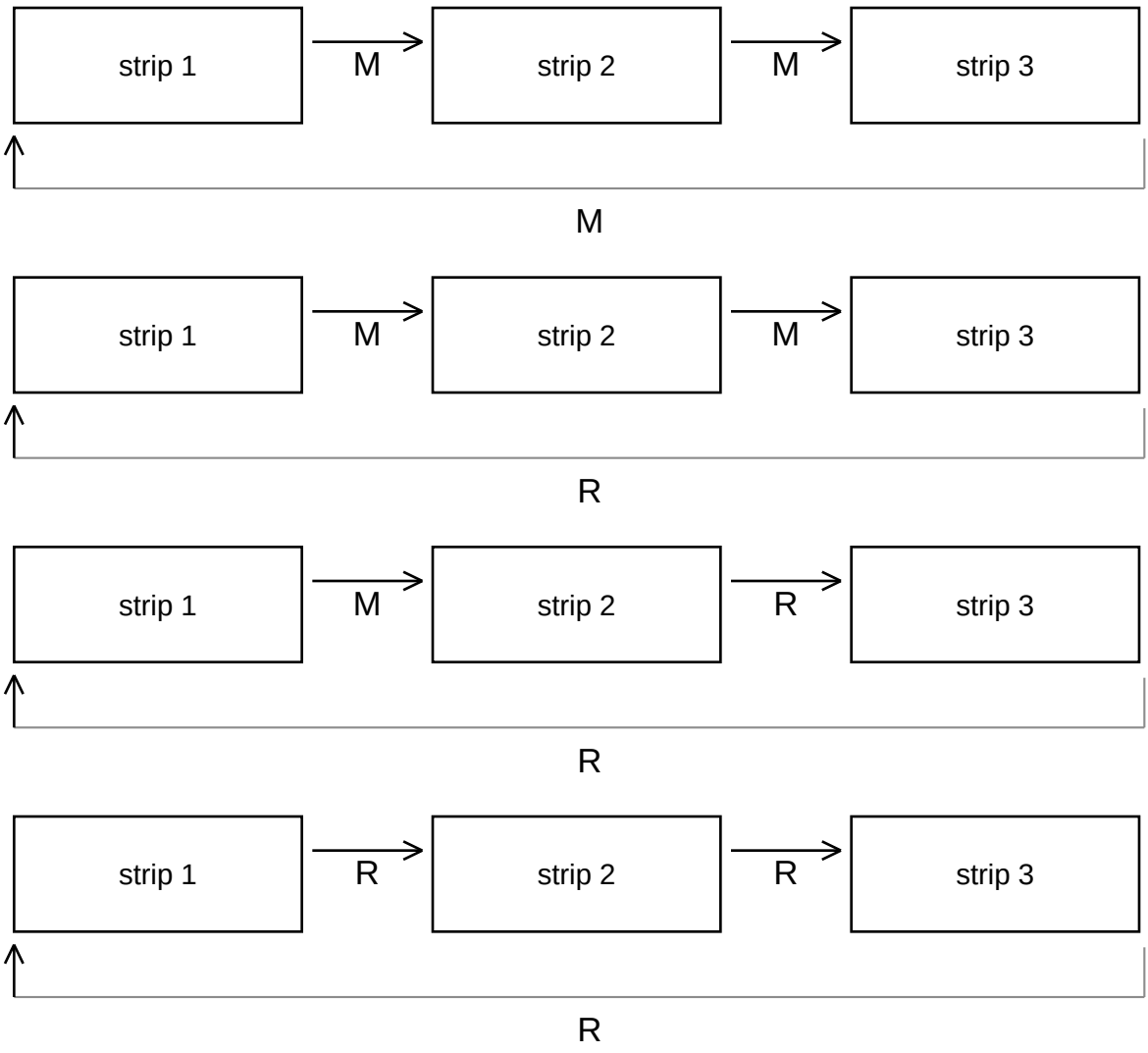


4 lanes / R

Join short ends; keep each mark on its corner. M joins dot to dot; R half-turns strip 2, then joins dot to square.

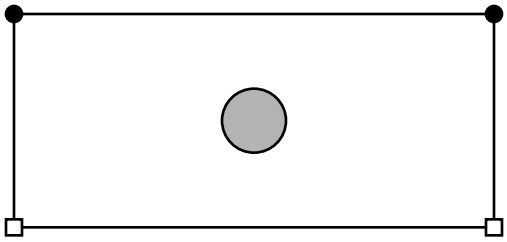


Problem 2: Join three strips into a circle using each seam recipe below. Find a rule predicting the number of closed edges. Design a four-strip circle with one closed edge, and another with two. Explain why the rule covers every choice of seams.

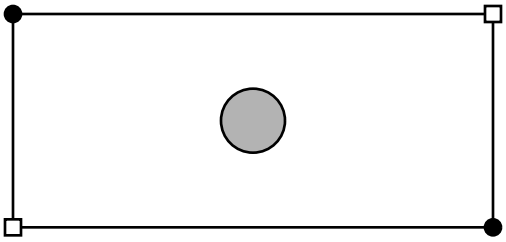


An adult removes the shaded paper before joining. Holes and notches stay small, separate, and narrower than the strip.

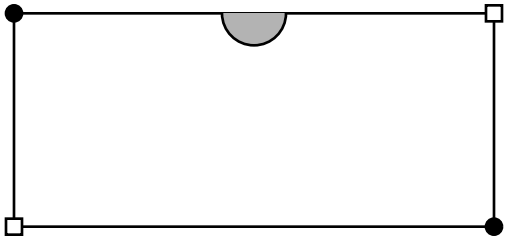
Problem 3: Trace every closed edge of bands A, B, C and D. How many does each have? Design a connected band with four closed edges using at most three holes.



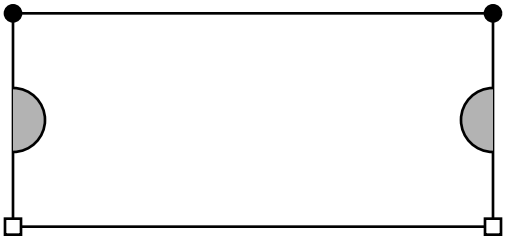
A / M



B / R



C / R



D / M